Practice 5 — Combinatorics (Basic Rules)

Solutions to Selected Problems

§1: Problems 3, 5, 10 §2: Problems 5, 10, 12

Discrete Mathematics — QYSU

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Section 1: Fundamental Rules (Sum and Product Rules)

1 Problem 3 — Choosing Two Books of Different Subjects

Theory Recap: Sum and Product Rules

The two most fundamental counting principles:

- **Sum Rule (Rule of Alternatives):** If a task can be performed in one of k mutually exclusive ways, and way i can be done in n_i ways, then the total number of ways is $n_1 + n_2 + \cdots + n_k$.
- **Product Rule (Rule of Sequential Choices):** If a task consists of k sequential steps, where step i can be performed in n_i ways (independently of the choices in other steps), then the total number of ways is $n_1 \cdot n_2 \cdots n_k$.

Combined usage. When a task can be decomposed into several *cases* (sum rule), each of which involves *sequential choices* (product rule), we compute the count for each case using the product rule, then add the results.

Problem Statement

In how many ways can you choose two books of different subjects if there are 15 books on Informatics, 12 on Mathematics, and 10 on Physics?

Solution

Since the two chosen books must be on *different* subjects, we enumerate all possible pairs of subjects. There are three cases (sum rule), and within each case we apply the product rule:

Case	Subject 1	Subject 2	Count
Informatics + Mathematics	15 choices	12 choices	$15 \cdot 12 = 180$
Informatics + Physics	15 choices	10 choices	$15 \cdot 10 = 150$
Mathematics + Physics	12 choices	10 choices	$12 \cdot 10 = 120$

By the sum rule (the three cases are mutually exclusive), the total is:

$$15 \cdot 12 \;+\; 15 \cdot 10 \;+\; 12 \cdot 10 \;=\; 180 + 150 + 120 \;=\; 450.$$

There are $\boxed{450}$ ways to choose two books of different subjects.

Alternative (complement method). Total handshakes among all $15 + 12 + 10 = 37$ people: $\binom{37}{2}$. Subtract handshakes within each group (same subject): $\binom{15}{2} + \binom{12}{2} + \binom{10}{2}$. Result: $\binom{37}{2} - \binom{15}{2} - \binom{12}{2} - \binom{10}{2} = 666 - 105 - 66 - 45 = 450$. Both methods agree. ✓

2 Problem 5 — Natural Numbers Less Than 700 Divisible by 5 and/or 3

Theory Recap: Counting Multiples in a Range

The number of positive multiples of d that are less than or equal to N is $\left\lfloor \frac{N}{d} \right\rfloor$. More precisely, the multiples of d in $\{1, 2, \ldots, N\}$ are $d, 2d, 3d, \ldots, \lfloor N/d \rfloor \cdot d$, so there are exactly $\lfloor N/d \rfloor$ of them.

"Less than N" vs. "at most N". The natural numbers less than N are $\{1, 2, \ldots, N-1\}$. The number of multiples of d among these is $\left\lfloor \frac{N-1}{d} \right\rfloor$.

Problem Statement

How many natural numbers less than 700 are divisible by 5? How many of those are divisible by 3? How many are divisible by both 3 and 5?

Solution

We consider the set $S = \{1, 2, 3, \ldots, 699\}$ (natural numbers less than 700).

(a) Divisible by 5. The multiples of 5 in S are $5, 10, 15, \ldots, 695$. The number of such multiples is:

$$\left\lfloor \frac{699}{5} \right\rfloor = \lfloor 139.8 \rfloor = 139.$$

(b) Divisible by 3 (among natural numbers less than 700). (Here "of those" refers to all natural numbers less than 700, not just the multiples of 5 from part (a).) The multiples of 3 in S are $3, 6, 9, \ldots, 699$. The number of such multiples is:

$$\left\lfloor \frac{699}{3} \right\rfloor = 233.$$

(c) Divisible by both 3 and 5. A number divisible by both 3 and 5 is divisible by $\operatorname{lcm}(3, 5) = 15$. The multiples of 15 in S are $15, 30, 45, \ldots, 690$. The number of such multiples is:

$$\left\lfloor \frac{699}{15} \right\rfloor = \lfloor 46.6 \rfloor = 46.$$

Divisible by 5: $\boxed{139}$. Divisible by 3: $\boxed{233}$. Divisible by both 3 and 5: $\boxed{46}$.

3 Problem 10 — 7-Digit Numbers with Alternating Parity

Theory Recap: Product Rule for Digit Counting

When counting numbers with digit-by-digit constraints, the product rule is applied position by position.

Key consideration: The *leading digit* (leftmost position) of an n-digit number **cannot be 0**, since that would reduce the number of digits. This restriction must always be accounted for separately.

Even digits: $\{0, 2, 4, 6, 8\}$ — five digits.
Odd digits: $\{1, 3, 5, 7, 9\}$ — five digits.

Problem Statement

Find the number of 7-digit numbers where the even-numbered positions contain odd digits and the odd-numbered positions contain even digits.

Solution

We label positions $1, 2, 3, 4, 5, 6, 7$ from left to right.

Pos. 1	Pos. 2	Pos. 3	Pos. 4	Pos. 5	Pos. 6	Pos. 7
odd (position)	*even* (position)	*odd* (position)	*even* (position)	*odd* (position)	*even* (position)	*odd* (position)
even digit	odd digit	even digit	odd digit	even digit	odd digit	even digit

Even positions (2, 4, 6): These must contain **odd** digits from $\{1, 3, 5, 7, 9\}$. Each position has 5 choices, giving 5^3 combinations.

Odd positions (1, 3, 5, 7): These must contain **even** digits from $\{0, 2, 4, 6, 8\}$. Each position has 5 choices *in general*, **except** position 1 (the leading digit), which **cannot be 0**.

- Position 1: $\{2, 4, 6, 8\}$ — **4 choices**.
- Positions 3, 5, 7: $\{0, 2, 4, 6, 8\}$ — 5 choices each, giving 5^3.

Applying the product rule. Since the choices at each position are independent:

$$\underbrace{4}_{\text{pos. 1}} \times \underbrace{5^3}_{\text{pos. 2, 4, 6}} \times \underbrace{5^3}_{\text{pos. 3, 5, 7}} = 4 \times 125 \times 125 = 4 \times 15{,}625 = 62{,}500.$$

Example. 2 143 658 is a valid 7-digit number (leading digit $\neq 0$). But 0 143 658 is not, because it would be a 6-digit number. The leading-digit restriction removes exactly $\frac{1}{10}$ of all 7-character strings.

There are $\boxed{62{,}500}$ such 7-digit numbers.

Section 2: Permutations and Arrangements

4 Problem 5 — Boys and Girls in a Line

Theory Recap: Permutations with Positional Constraints

When certain positions in a line are constrained (e.g., "the first and last positions must be occupied by specific types of people"), we use the following strategy:

1. **Fill the constrained positions first**, counting the number of ways to choose who goes there.
2. **Fill the remaining positions** with the remaining people, who can stand in any order.
3. **Multiply** the counts (product rule).

Recall that the number of ways to arrange n distinct objects in a line is $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$.

Problem Statement

In how many ways can 6 boys and 7 girls stand in a line such that the first and last positions are occupied by boys?

Solution

There are $6 + 7 = 13$ people in total.

$$\boxed{B}\ _\ _\ _\ \cdots\ _\ _\ \boxed{B}$$

Step 1: Fill the constrained positions.

- **First position:** must be a boy. There are 6 boys to choose from, so 6 choices.
- **Last position:** must be a boy (different from the first). There are $6 - 1 = 5$ remaining boys, so 5 choices.

Step 2: Fill the remaining positions. After placing two boys at the ends, there are $13 - 2 = 11$ people left (4 boys and 7 girls). These 11 people can stand in the remaining 11 positions (positions 2 through 12) in any order:

$$11! \text{ ways.}$$

Step 3: Apply the product rule. The choices at each step are independent, so:

$$6 \times 5 \times 11! \ = \ 30 \times 11!.$$

The total number of arrangements is $\boxed{30 \times 11! = 30 \times 39{,}916{,}800 = 1{,}197{,}504{,}000}$.

5 Problem 10 — Ramsey's Theorem $R(3,3) = 6$

Theory Recap: Pigeonhole Principle and Ramsey Theory

Pigeonhole Principle. If n objects are placed into k boxes and $n > k$, then at least one box contains more than one object.
More generally: if n objects are placed into k boxes, then at least one box contains $\lceil n/k \rceil$ objects.

Ramsey Theory (informal statement). In any sufficiently large structure, order must appear. The Ramsey number $R(s,t)$ is the smallest n such that every 2-colouring (say, red and blue) of the edges of the complete graph K_n contains either a red K_s or a blue K_t.

The classical result $R(3,3) = 6$ states: in any group of 6 people, there are always 3 mutual friends or 3 mutual strangers.

Problem Statement

Prove that in any group of 6 people, there are either 3 mutual friends or 3 mutual strangers.

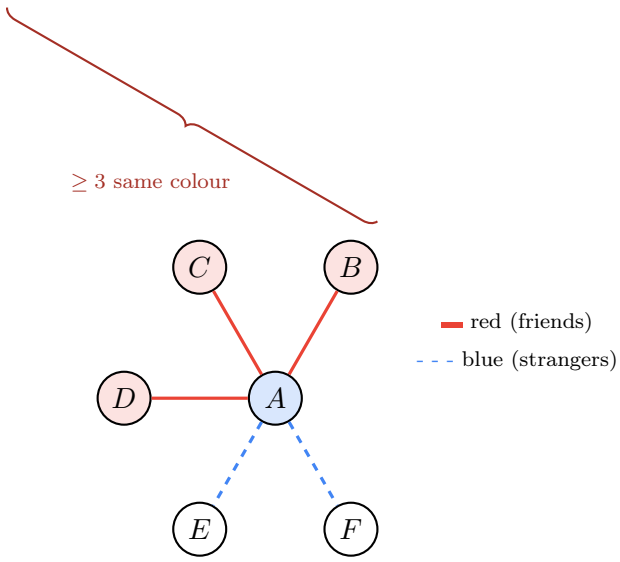
Solution (Proof)

We model the situation as a 2-colouring of the complete graph K_6. Each person is a vertex, and each edge is coloured **red** (friends) or **blue** (strangers). We must show that there exists a monochromatic triangle.

Step 1: Apply the Pigeonhole Principle. Pick any vertex A. Vertex A is connected to the other 5 vertices by 5 edges, each coloured red or blue. By the pigeonhole principle (5 edges into 2 colours), at least $\lceil 5/2 \rceil = 3$ edges from A have the same colour.

Without loss of generality, assume A has at least 3 **red** edges. Let the other endpoints of these red edges be B, C, D. (The case with 3 blue edges is handled symmetrically by swapping "friend" and "stranger.")

Why WLOG works. If vertex A instead has 3 blue edges, repeat the exact same argument with the roles of blue and red swapped. The problem is symmetric in the two colors, so the conclusion is the same.



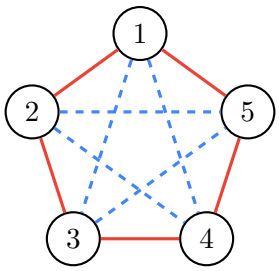
Step 2: Examine the edges among B, C, D. Consider the three edges BC, BD, CD. There are two cases:

Case 1: At least one of BC, BD, CD is **red**.
Suppose (WLOG) BC is red. Then A, B, C form a red triangle: edges AB, AC, and BC are all red. Thus we have 3 mutual **friends**.

Case 2: None of BC, BD, CD is red, i.e., all three are **blue**.
Then B, C, D form a blue triangle: edges BC, BD, CD are all blue. Thus we have 3 mutual **strangers**.

In both cases, a monochromatic triangle exists. □

Why 5 people are not enough. To show that $R(3,3) > 5$, we exhibit a 2-colouring of K_5 with no monochromatic triangle: arrange 5 vertices in a regular pentagon and colour the edges of the pentagon red and the diagonals blue (or vice versa). One can verify that no triangle is monochromatic.



K_5: edges red, diagonals blue — no monochromatic triangle

In any group of 6 people, there are always 3 mutual friends or 3 mutual strangers.
This proves $R(3,3) = 6$.

6 Problem 12 — Divisibility Among $7, 77, 777, \ldots$

Theory Recap: Pigeonhole Principle for Divisibility

A powerful application of the pigeonhole principle in number theory:

If we have $n+1$ integers and consider their remainders modulo n, then by the pigeonhole principle, at least two of them share the same remainder. Their *difference* is therefore divisible by n.

Repunit-like sequences. The sequence $7, 77, 777, 7777, \ldots$ can be written in closed form. The k-th term is:

$$a_k = \underbrace{77\ldots7}_{k \text{ sevens}} = 7 \cdot \underbrace{11\ldots1}_{k} = 7 \cdot \frac{10^k - 1}{9}.$$

Problem Statement

Prove that among the first 2014 terms of the sequence 7, 77, 777, 7777, ... there is one divisible by 2013.

Solution (Proof)

Let $a_k = \underbrace{77\ldots7}_{k}$ for $k = 1, 2, \ldots, 2014$.

Consider the 2014 remainders $r_k = a_k \bmod 2013$ for $k = 1, 2, \ldots, 2014$.

Case 1: Some $r_k = 0$. If $r_k = 0$ for some k, then a_k is divisible by 2013, and we are done.

Case 2: No $r_k = 0$. Then each $r_k \in \{1, 2, \ldots, 2012\}$. We have 2014 remainders taking values in a set of size 2012. By the **pigeonhole principle**, there exist indices $i < j$ (with $1 \le i < j \le 2014$) such that $r_i = r_j$, i.e.,

$$a_j \equiv a_i \pmod{2013}.$$

Therefore $2013 \mid (a_j - a_i)$.

Analysing $a_j - a_i$. We compute:

$$a_j - a_i = \underbrace{77\ldots7}_{j} - \underbrace{77\ldots7}_{i} = \underbrace{77\ldots7}_{j-i} \cdot 10^i = a_{j-i} \cdot 10^i.$$

So $2013 \mid a_{j-i} \cdot 10^i$.

Example. $\underbrace{77\ldots7}_{5} - \underbrace{77\ldots7}_{3} = 77777 - 777 = 77000 = 77 \times 1000 = \underbrace{77\ldots7}_{2} \times 10^3$.

Now, $2013 = 3 \times 11 \times 61$. Since $\gcd(10, 2013) = 1$ (as 2013 has no factors of 2 or 5), it follows that $\gcd(10^i, 2013) = 1$.

Therefore $2013 \mid a_{j-i}$.

Since $1 \leq j - i \leq 2013 < 2014$, the term a_{j-i} is indeed among the first 2014 terms of the sequence.

In both cases, there exists a term in $\{a_1, a_2, \ldots, a_{2014}\}$ that is divisible by 2013. $\square$

By the pigeonhole principle, among $a_1, a_2, \ldots, a_{2014}$, there is at least one term divisible by 2013.